Preparation

On the exam, you may be asked to do the following:

- Rephrase a key definition and/or theorem in your own words.
- Determine whether a given statement is true or false.
- Interpret or explain a statistics concept in everyday language.
- Perform calculations using relevant techniques from the course.
- Provide a short, rigorous proof of a novel statement or result.
- Create and analyze a statistical model for a particular phenomenon.

There are some practice problems below. These are by no means exhaustive!

Practice Problems

1. Let $X_1, \ldots, X_n|\theta \overset{\text{iid}}{\sim} f(x|\theta)$ where for $x \in \mathbb{R}$,

$$f(x|\theta) = \frac{1}{2\theta} e^{-|x|/\theta}$$

   where θ is an unknown parameter with $\theta > 0$. Find an MLE for θ.

2. Let $X_1, \ldots, X_n|\theta \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$. We have two candidate estimators of θ: $\delta_1 = \bar{X}$ and $\delta_2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Which estimator is better, and why?

3. Suppose that the random variable X has a binomial distribution with unknown value n and a known value of p for $0 < p < 1$. Determine the MLE of n based on the observation X.

4. Suppose a single observation X is taken from a $N(0, \sigma^2)$ distribution.

   (a) Find a method of moments estimator of σ.

   (b) Is your estimator from (a) sufficient for σ?

   (c) What is the bias of your estimator from (a) for σ?

5. The method of *randomized response* is sometimes used to conduct surveys on sensitive topics. A simple version of the method can be described as follows:

   - A random sample of n people are drawn from a large population.
   - For each person in the sample, there is probability 1/2 that the person will be asked a standard question and probability 1/2 that the person will be asked a sensitive question. Furthermore, this selection of the standard or sensitive question is made independently from person to person.

- If a person is asked the standard question, then there is probability 1/2 that the person will give a positive response; however, if the person is asked the sensitive question, then there is an unknown probability p that they will give a positive response.

The statistician can observe only the total number X of positive responses that were given by the n people in the sample, but cannot observe who nor how many were asked the sensitive question.

Determine the MLE of p based on the observation X.

6. Consider a sequence of independent binary trials $X_1, X_2, \ldots$ each with probability θ of success. Let N_n be the number successes in the first n trials (i.e. $N_n = \sum_{i=1}^{n} X_i$). Let T_r be the trial in which the r-th success occurs, for $r \geq 2$.

   (a) Demonstrate that $P(T_r = k) = P(N_k - 1 = r - 1)\theta$.

   (b) In words, explain why $N_{T_r - 1} = r - 1$. Also in words, explain what $N_{T_r-1}/(T_r - 1)$ represents.

   (c) Show that $(r-1)/(T_r - 1)$ is an unbiased estimator of θ. *Hint: consider Law of Iterated Expectation.*

   (d) Is r/T_r is a biased estimator of θ? If so, does this estimator tend to under- or over-estimate θ? Why do you think that is? *Hint: add 0 to r/T_r to sneak in your unbiased estimator and then manipulate terms!*

7. Suppose $X|\theta \sim \text{Binom}(n, \theta)$ where the success probability $\theta \in (0, 1)$ is unknown. Suppose we are interested in estimating $g(\theta) = \Pr(X = 1)$, the probability of a single success in this Binomial observation. I propose the estimator $\delta(X) = \mathbf{1}_{\{X=1\}}$.

   (a) Is my estimator $\delta(X)$ an unbiased estimator for the quantity of interest?

   (b) Find MSE $(\delta(X), g(\theta))$.

8. Provide three reasons why the Maximum Likelihood Estimator might be the default/preferred choice of estimator. When might you prefer an Method of Moments estimator over the MLE?

9. Suppose you are assessing the rate of radioactive decay for three different isotopes of an unknown element. The amount of time X_1 between particle emissions for the **first** isotope is exponential with mean θ, the amount of time X_2 between particle emissions for the **second** isotope is exponential with mean 2λ, and the amount of time X_3 between particle emissions for the **third** isotope is exponential with mean $3\alpha\lambda$. Assume that $\alpha > 1, \lambda > 0$, and that given the values of α and λ, the amount of time between particle decays are independent for the three isotopes. Let $\boldsymbol{\theta} = (\lambda, \alpha)$ be the unknown parameters.

   (a) Give an explicit description of the statistical model by identifying the joint probability distribution of the data and by identifying the parameters and the parameter space.

(b) Calculate the MLE of $\boldsymbol{\theta}$ based on the observations $X_1 = x_1$, $X_2 = x_2$ and $X_3 = x_3$. *You technically should justify that the critical points of the likelihood function correspond to a local maximum, but in this problem the Hessian is not beautiful so don't worry about it.*

10. Suppose $X_1, \ldots, X_n | \theta \overset{\text{iid}}{\sim} \text{Exp}(\theta)$. Find an efficient estimator for some function $g(\theta)$ of your choice! Can you obtain this result using two different approaches?

11. Suppose $X_1, \ldots, X_n | \boldsymbol{\theta} \overset{\text{iid}}{\sim} \text{Binom}(m, p)$ where $\boldsymbol{\theta} = (m, p)$ are unknown. Find a method of moments estimator for $\boldsymbol{\theta}$.